

Student Success Intelligence Framework: Longitudinal Survival Trajectories, Academic Resilience, and Cross-Study Digital Telemetry Governance

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Abstract—Student attrition and post-graduation employability represent critical challenges in higher education worldwide. Traditional student retention models treat semester observations cross-sectionally, introducing severe data leakage while failing to capture cumulative momentum. In this paper, we present the Student Success Intelligence Framework (SSIF), an open-source, mathematically audited computational framework evaluated on 79,239 longitudinal student-semester observations ($N = 20,000$ students) and an MBA employability cohort ($N = 215$).

First, we design a vectorized, closed-form ordinary least squares (OLS) trajectory engine operating with strict temporal causality ($\leq t$), computing cumulative GPA slopes, volatility, and decline indices in 0.15s. Second, evaluated under 5-Fold GroupKFold cross-validation (grouped by student), our multi-tier models achieve an AUROC of 0.8014 and PR-AUC of 0.3643 (vs. 0.0860 baseline prevalence). Third, Kaplan-Meier and Cox Proportional Hazards modeling ($C = 0.7498$) reveal that first-generation students face nearly double instantaneous departure hazard ($HR = 1.98, p < 0.001$), while institutional scholarships reduce hazard by 48% ($HR = 0.52$). Fourth, unsupervised K-Means clustering ($k = 3$) with bootstrap stability (Adjusted Rand Index = 0.9703) discovers three persistent phenotypes, identifying an acute collapse cohort with a 60.3% dropout rate. Analysis of $N = 5,563$ recovering students proves that proactive academic advising increases odds of academic rebound by +73.1% per visit ($OR = 1.731, p < 0.001$), whereas financial stress forms the primary structural barrier ($OR = 0.666$). Finally, an empirical feature ablation study ($\Delta AUROC = -0.00005, p = 0.932$) rigorously disproves naive row-level integration with digital lifestyle telemetry (DLSM), while establishing a validated representation-level construct bridge.

Index Terms—Educational Data Mining, Student Retention, Survival Analysis, Longitudinal Trajectories, Algorithmic Recourse, Cross-Study Governance.

1 Introduction

Higher education institutions face compounding attrition rates that disproportionately affect socio-economically vulnerable student cohorts. Standard early-warning predictive systems suffer from three pervasive limitations:

- 1) **Cross-Sectional Naivety:** Classifiers evaluate semester snapshots independently, ignoring the velocity and historical momentum of grade decay.
- 2) **Target Contamination & Leakage:** Random cross-validation splits allow observations from the same student across semesters 1 through 8 to contaminate both training and validation folds, producing artificially optimistic generalization metrics.
- 3) **Unjustified Multi-Dataset Merges:** Researchers frequently force row-level concatenations between disparate educational and digital telemetry datasets without testing empirical variable overlap or demographic alignment.

To resolve these challenges, we formalize the Student Success Intelligence Framework (SSIF), providing rigorous mathematical formulations for longitudinal trajectory dy-

namics, time-to-event survival hazard estimation, academic resilience analysis, and cross-study compatibility gating.

2 Longitudinal Trajectory Engine

Let student $i \in \{1, \dots, M\}$ be observed across semesters $s \in \{1, \dots, t_i\}$. To prevent temporal leakage (RULE-009), all features at semester t must be measurable strictly on the historical filtration $\mathcal{F}_{i,t} = \{(\text{Semester}_s, \text{GPA}_s) : s \leq t\}$.

2.1 Closed-Form OLS Slope Formulation

The cumulative linear GPA trajectory slope for student i at semester $t \geq 2$ is computed via vectorized summation:

$$\beta_{i,t} = \frac{n \sum_{s=1}^t s \cdot Y_{i,s} - \left(\sum_{s=1}^t s\right) \left(\sum_{s=1}^t Y_{i,s}\right)}{n \sum_{s=1}^t s^2 - \left(\sum_{s=1}^t s\right)^2} \quad (1)$$

where $Y_{i,s} = \text{Sem_GPA}_{i,s}$ and $n = t$.

2.2 Decline and Recovery Indices

Let GPA velocity be $v_{i,t} = Y_{i,t} - Y_{i,t-1}$. The consecutive decline index $D_{i,t}$ is the cumulative run-length of negative velocity:

$$D_{i,t} = \begin{cases} D_{i,t-1} + 1 & \text{if } v_{i,t} < 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

The binary recovery signature $R_{i,t}$ indicates an upward inflection following a dip:

$$R_{i,t} = \mathbb{I}(v_{i,t} > 0 \wedge v_{i,t-1} < 0) \quad (3)$$

H. G. is an independent machine learning and computational education researcher. Project repository: <https://github.com/HarshkumarG007/SSIF>. Related project: Digital Lifestyle Spillover Modeling (DLSM), <https://github.com/HarshkumarG007/DLSM>. Contact: <https://dlsm-research.streamlit.app/>.

3 Predictive Benchmark & Survival Modeling

3.1 GroupKFold Benchmark Results

Retention models are validated using 5-Fold GroupKFold partitioning grouped on Student_ID. As shown in Table 1, the trajectory-augmented Logistic Regression achieves superior calibration and discrimination.

TABLE 1
Cross-Validated Model Performance (5-Fold GroupKFold)

Model Tier	AUROC	PR-AUC	Brier	ECE
Tier 0: Majority Baseline	0.4945	0.0860	0.0797	0.0000
Tier 1: Logistic Regression	0.8013	0.3642	0.0669	0.1768
Tier 1+: LogReg + Traj.	0.8014	0.3643	0.0669	0.1768
Tier 3: Random Forest	0.7874	0.3223	0.1177	0.2002
Tier 4+: HistGBM + Traj.	0.7975	0.3521	0.1692	0.2857

3.2 Cox Proportional Hazards Analysis

Survival modeling treats graduated or currently enrolled students as right-censored. The semi-parametric hazard rate is formalized as:

$$h(t | \mathbf{x}) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{x}) \quad (4)$$

The model achieves Harrell’s Concordance Index $C = 0.7498$ ($p < 0.001$).

- First-Generation Hazard Ratio: $HR = 1.98$ (95% CI: [1.89, 2.08]), indicating that first-generation students face nearly double instantaneous departure risk.
- Scholarship Protection: $HR = 0.52$ (95% CI: [0.49, 0.55]), reducing hazard by 48%.
- Academic Performance: $HR = 0.40$ per unit GPA increase.

4 Trajectory Phenotypes & Resilience

4.1 Cluster Stability (RULE-017)

Applying K-Means ($k = 3$) to trajectory dynamics yields exceptional bootstrap stability across $B = 15$ resamples with Adjusted Rand Index (ARI) of 0.9703.

- 1) Precipitous Collapse (17.7%): Mean slope $-0.37/\text{sem}$; 60.3% eventual dropout rate.
- 2) Chronic Erosion (22.5%): Continuous consecutive decline index of 4.17 sems; 28.4% dropout rate.
- 3) Stable Persistence (59.8%): Low volatility (0.13); 8.1% dropout rate.

4.2 Resilience Analysis ($N = 5,563$)

Among students experiencing severe grade decline ($\Delta\text{GPA} \leq -0.3$), $N = 5,563$ achieved academic recovery. The recovery cohort achieved a 22.6% dropout rate vs. 41.9% for unrecovered peers ($p < 10^{-50}$). Multivariate logistic regression reveals that academic advising is the most powerful institutional intervention ($OR = 1.731$, $p < 0.001$), while financial stress forms the dominant impediment ($OR = 0.666$, $p < 0.001$).

5 Cross-Study DLSS Governance & Feature Ablation

We empirically evaluated integration feasibility with the Digital Lifestyle Spillover Modeling (DLSSM) dataset.

- Variable Overlap Score: 0.154 (only Age and Gender compatible; core sleep and screen variables absent).
- 5-Fold GroupKFold Feature Ablation: Comparing Academic-only baseline (A_0) against Academic + DLSSM demographics (A_1):

$$\text{AUROC}(A_0) = 0.80130 \pm 0.0052$$

$$\text{AUROC}(A_1) = 0.80125 \pm 0.0052$$

$$\Delta\text{AUROC} = -0.00005 \quad (t = -0.089, p = 0.932)$$

Conclusion: Direct row-level merging offers zero incremental predictive value. However, a representation-level construct bridge (Wasserstein distance = 1.767 years) proves that digital cognitive fatigue and academic strain represent parallel vulnerability pathways.

6 Algorithmic Counterfactual Recourse

For students flagged at high departure risk, academic advisors require actionable intervention prescriptions. Given student profile $\mathbf{x}$, target risk $\tau = 0.15$, and actionable feature set $\mathcal{A} \subset \{1, \dots, p\}$, the recourse optimization problem is:

$$\min_{\mathbf{x}^*} \sum_{j \in \mathcal{A}} c_j \left| \frac{x_j^* - x_j}{\sigma_j} \right| \quad \text{s.t.} \quad P(\text{Dropout} | \mathbf{x}^*) \leq \tau \quad (5)$$

where immutable demographic attributes remain strictly fixed.

7 Conclusion

The Student Success Intelligence Framework proves that rigorous longitudinal trajectory engineering, survival hazard modeling, and algorithmic counterfactual recourse enable proactive student retention. Crucially, empirical ablation establishes that digital habit integration must occur through synchronous prospective telemetry rather than manufactured post-hoc merges.

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